Status	Finished
Started	Tuesday, 20 October 2026, 6:07 PM
Completed	Tuesday, 20 October 2026, 6:10 PM
Duration	3 mins 14 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Decreases the efficiency of algorithms
c. Makes data harder to manage
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **2**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Quick Sort
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Quick sort
- c. Insertion sort
- d. Heap sort

The correct answer is:
Insertion sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It divides the list into two halves, sorts each half, and then merges them
- c. It compares adjacent elements and swaps them if necessary
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **5**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. sorted()
- c. arrange()
- d. order()

The correct answer is: sort()

Question **6**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 200,470,80,150,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Searching
- c. Merging
- d. Rearranging

The correct answer is: Sorting

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It only works with integer arrays
- c. It requires external libraries
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. arrange()
- c. order()
- d. sorted()

The correct answer is: sorted()

Question **10**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are combined to form a single sorted list
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **12**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Searching

c. Merging

d. Complexity

The correct answer is:
Complexity

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement

b. Bubble Sort cannot handle duplicate elements

c. Bubble Sort does not guarantee sorted order

d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others

b. Sorting the entire list sequentially

c. Combining sorted sublists

d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **15**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. The parts are merged without sorting
c. Each part is searched for a specific element
d. The parts are ignored

The correct answer is: Each part is sorted independently